

AI Feature Pipelines: Traditional ML vs Generative AI

■■ Tuesday, Sept 22, 2026: The Breakdown (The Concept Simply Explained)

If you want to manage AI feature pipelines effectively, you have to establish a strict line between two completely distinct systems.

Let's strip away the vendor marketing jargon. The AI landscape splits into two main engines:

1. Traditional Machine Learning (The Predictor)

- **How it works:** It uses statistical math and algorithms to find patterns in structured data.
- **Nature:** It is **deterministic**. If you feed it the exact same input payload, it calculates the exact same mathematical probability score every single time.
- **Best for:** Classification, forecasting, and analytical scores (e.g., predicting user churn, scoring credit risk, or sorting support tickets).

2. Generative AI (The Creator)

- **How it works:** It uses deep neural networks to predict the next most likely token or pixel based on a massive training dataset.
- **Nature:** It is **probabilistic**. It doesn't calculate a fixed answer; it dynamically constructs a brand-new response string.
- **Best for:** Unstructured text, summarisation, creative coding assistance, and linguistic reasoning.

Think of it this way:

- Traditional ML **analyzes** what already exists.
- Generative AI **creates** something new.

Tomorrow, I'll share a real-world case study where mixing these two concepts up inside production completely broke an e-commerce application.

CTA: Do you find your product stakeholders mixing these two concepts up when requesting engineering roadmaps?



1. Traditional Machine Learning (The Predictor)



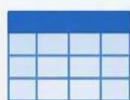
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Structured Data



ML Model



Probability Score
(Always the same)



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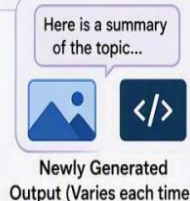
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Prompt / Input



Generative Model
(LLM / Diffusion Model)



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